A Complete Projective Atlas for Spheroidal Jeffery Directors in Planar Linear Flow

HCS Research Program

1 September 2026

Abstract

We classify a passive axisymmetric spheroid in every steady incompressible planar linear flow. The nonlinear head–tail director equation on $\mathbb{RP}^2$ is the projectivization of one traceless matrix exponential. A scalar discriminant then yields the complete elliptic, hyperbolic, nilpotent, and identity atlas. We prove the full source–saddle–sink stratification, the nilpotent fixed projective line, and the true head–tail versus oriented simple-shear periods. Every stroboscopic fixed set and every sphere, rod, disk, and zero-flow boundary is included. The result is source-local: clean periodic continua supply no arithmetic primitive-orbit owner or target determinant.

1 Frozen director and its linear lift

Let

$$L = \begin{pmatrix} a & b & 0 \\ c & -a & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad E = \tfrac{1}{2}(L + L^T), \quad W = \tfrac{1}{2}(L - L^T).$$

For a spheroid of finite aspect ratio $r > 0$, put

$$\lambda = \frac{r^2 - 1}{r^2 + 1} \in (-1, 1). \tag{1}$$

The head–tail director is $[p] \in \mathbb{RP}^2$. A unit representative obeys

$$\dot{p} = Wp + \lambda\{Ep - (p^T Ep)p\}. \tag{2}$$

For $r \neq 1$, $[p]$ is the intrinsic symmetry-axis director. An unmarked sphere ($r = 1$) has no intrinsic shape director; whenever $r = 1$ appears below, $[p]$ denotes a marked material director carried by the sphere. Jeffery [1] is cited only for the isolated-ellipsoid source equation, and Bretherton [2] only for rigid-particle and spheroidal-parameter lineage. No classification or proof below is outsourced.

Set $B = W + \lambda E$. Because $p^T Wp = 0$, (2) is $\dot{p} = Bp - (p^T Bp)p$.

Lemma 1 (Exact projective lift)**.** *For every nonzero $q_0 \in \mathbb{R}^3$,*

$$[p(t)] = [e^{tB} q_0], \qquad p(t) = \frac{e^{tB} q_0}{\|e^{tB} q_0\|}. \tag{3}$$

Conversely every solution of (2) *has this form, so the flow is global on $\mathbb{RP}^2$.*

Proof. If $q' = Bq$ and $p = q/\|q\|$, differentiation gives $p' = Bp - (p^T Bp)p$. Conversely, multiplying a unit solution by $\exp \int_0^t p^T Bp \, ds$ gives a solution of $q' = Bq$. Matrix exponentials are invertible, so normalization never vanishes. Passing to $[p]$ removes the representative sign exactly. □

Write $k = (b - c)/2$ and $s = (b + c)/2$. The active block and its invariant are

$$B_2 = \begin{pmatrix} \lambda a & k + \lambda s \\ -k + \lambda s & -\lambda a \end{pmatrix}, \qquad \delta = \lambda^2(a^2 + s^2) - k^2. \tag{4}$$

Direct multiplication gives $B_2^2 = \delta I$ and $\delta = -\det B_2$.

2 The complete sign atlas

Theorem 1 (All projective orbit types)**.** *The sign and rank in (4) give exactly four cases.*

1. *If $\delta < 0$ and $\omega = \sqrt{-\delta}$, the vertical director is the only fixed point. Every equatorial director has least period π/ω; every director with nonzero horizontal and vertical parts has least period $2\pi/\omega$.*
2. *If $\delta > 0$ and $\rho = \sqrt{\delta}$, there are exactly three fixed eigen-directors $[v_+]$, $[e_0]$, $[v_-]$ with exponents $\rho, 0, -\rho$. Writing $q = q_+v_+ + q_0e_0 + q_-v_-$, its forward limit is $[v_+]$ if $q_+ \neq 0$, $[e_0]$ if $q_+ = 0, q_0 \neq 0$, and $[v_-]$ otherwise; backward time reverses $+$ and $-$. Thus they are sink, saddle, and source. Precisely,*
$$W^s([e_0]) = \mathbb{P}\,\mathrm{span}(e_0, v_-) \setminus \{[v_-]\}, \qquad W^u([e_0]) = \mathbb{P}\,\mathrm{span}(e_0, v_+) \setminus \{[v_+]\}.$$
Their closures are the two invariant $\mathbb{RP}^1$ projective lines.
3. *If $\delta = 0$ but $B_2 \neq 0$, the fixed set is the entire projective line $\mathbb{P}(\ker B)$. Every other trajectory converges as $t \to \pm\infty$ to the unique horizontal line $\mathbb{P}(\mathrm{im}\, B_2) = \mathbb{P}(\ker B_2)$, at algebraic rate $|t|^{-1}$.*
4. *If $B_2 = 0$, the flow is the identity on all of $\mathbb{RP}^2$.*

Proof. Cayley–Hamilton gives, respectively,
$$e^{tB_2} = \cos(\omega t)I + \frac{\sin(\omega t)}{\omega}B_2, \qquad e^{tB_2} = \cosh(\rho t)I + \frac{\sinh(\rho t)}{\rho}B_2. \tag{5}$$

In the first case there is no real horizontal eigenline. At π/ω the horizontal block is $-I$, which returns every equatorial head–tail line but not a mixed line; at $2\pi/\omega$ the full matrix is I. No earlier return is possible because a non-scalar elliptic block has no real eigenline.

In the second case, exponentiating the three eigen-coordinates gives the stated limits. Within $\mathbb{P}\,\mathrm{span}(e_0, v_-)$, every point except $[v_-]$ tends forward to $[e_0]$; within $\mathbb{P}\,\mathrm{span}(e_0, v_+)$, every point except $[v_+]$ tends backward to $[e_0]$. These are projective lines, and a point outside them has both corresponding leading coordinates, proving the stable/unstable completeness. Coordinate ratios give gaps ρ or 2ρ. At $\delta = 0$, $e^{tB} = I + tB$. A nonzero nilpotent 2 by 2 block has rank one and image equal to kernel. Hence $\mathbb{P}(\ker B)$ is fixed, whereas $[q + tBq] \to [Bq]$ off that line. Finally $B_2 = 0$ and the zero vertical block give $B = 0$. □

3 Simple shear and singular shape faces

For $L_2 = \left(\begin{smallmatrix} 0 & \dot\gamma \\ 0 & 0 \end{smallmatrix}\right)$ with $\dot\gamma \neq 0$, substitution of (1) in (4) yields
$$\delta = -\frac{\dot\gamma^2 r^2}{(r^2+1)^2}, \qquad \omega = \frac{|\dot\gamma| r}{r^2+1}. \tag{6}$$

Thus the physical head–tail equatorial period and the oriented-vector period are
$$T_{\mathbb{RP}^1} = \frac{\pi(r + r^{-1})}{|\dot\gamma|}, \qquad T_{\text{oriented}} = \frac{2\pi(r + r^{-1})}{|\dot\gamma|}. \tag{7}$$

The second value is the least period of every nonvertical oriented vector and also of a mixed $\mathbb{RP}^2$ director; the vertical oriented vector is fixed. This factor two is a state-space fact, not a convention change.

At $r = 1$, $\lambda = 0$ and strain drops out: under the marked-material-director convention above, the marked direction follows only vorticity. The unmarked sphere itself has no observable orientation state. As $r \to \infty$ or $r \to 0$, $\lambda \to \pm 1$, the simple-shear generator is nilpotent and (7) diverges. These rod/disk limits are not finite-aspect periodic points. At $\dot\gamma = 0$, every director is fixed.

4 Every elliptic strobe

The first formula in (5) also resolves the return map without sampling.

Theorem 2 (Stroboscopic fixed sets). *For $\delta < 0$ and $\tau > 0$,*

$$\mathrm{Fix}_{\mathbb{RP}^2}(e^{\tau B}) = \begin{cases} \{[e_0]\}, & \omega\tau \notin \pi\mathbb{Z}, \\ \mathbb{RP}^1_{\mathrm{eq}} \cup \{[e_0]\}, & \omega\tau \in (2\mathbb{Z}+1)\pi, \\ \mathbb{RP}^2, & \omega\tau \in 2\pi\mathbb{Z}. \end{cases} \tag{8}$$

Proof. Away from integer half-periods the horizontal block has no real eigenline, so only the vertical eigenline fixes. At odd half-period it is $-I$ while the vertical multiplier is $+1$, fixing the two separate eigenspaces but no mixed line. At a full period every multiplier is one. □

Condition	Fixed set	Nonfixed limit	Recurrence
$\delta < 0$	vertical	none	clean periodic continua
$\delta > 0$	three lines	source/saddle/sink	none
$\delta = 0$, $B \neq 0$	one $\mathbb{RP}^1$	algebraic kernel limit	none
$B = 0$	all $\mathbb{RP}^2$	none	stationary

5 Executable receipt and claim boundary

The deterministic certificate contains 625 exact parameter rows, 320 90-digit orbit reconstructions, ten shear-period rows, five strobe rows, and six boundary rows. The producer-independent checker passes 8,328 assertions; the symbolic reconstruction passes 39 identities; replay is byte-exact; and 25/25 repaired-hash or stale-hash hostile mutations are rejected. These rows are regression oracles, not a finite proof of the arbitrary-parameter theorem.

Release item	Result
Parameter / orbit / shear / strobe / boundary rows	625/320/10/5/6
Independent assertions / symbolic identities	8,328/39 PASS
Fresh byte replay / hostile mutations	PASS / 25/25 rejected
Evidence SHA-256	`858a50dcdfc8…847efa308374`

The nearest projective, spin, and linear-Hamiltonian owners do not contain this passive $\mathbb{RP}^2$ shape/gradient discriminant. This workspace distinction is not a literature-priority claim.

The honest Route-A tuple is

$$(\text{A0\_FAIL, A1\_WEAK, A2\_FAIL, A3\_FAIL, A4\_FAIL}), \qquad \text{ROUTE\_A\_REJECTED}. \tag{9}$$

There is no rational-prime carrier or logarithmic clock. Elliptic periodic points occur in clean continua with continuously variable periods; the other chambers align or are stationary. Consequently no isolated primitive-orbit product, target determinant, target zero match, Hilbert–Pólya operator, or Route-B input follows. The locked scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

References

[1] G. B. Jeffery, "The motion of ellipsoidal particles immersed in a viscous fluid," *Proc. R. Soc. Lond. A* **102** (715), 161–179 (1922), doi:10.1098/rspa.1922.0078.

[2] F. P. Bretherton, "The motion of rigid particles in a shear flow at low Reynolds number," *J. Fluid Mech.* **14** (2), 284–304 (1962), doi:10.1017/S002211206200124X.